

Replication of Decarolis (2015), Table 8

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1 Paper Information

This report replicates Table 8 from [Decarolis \(2015\)](#), “Medicare Part D: Are Insurers Gaming the Low Income Subsidy Design?” The paper was published in the *American Economic Review*, volume 105, issue 4, pages 1547–1580.

The target result is Table 8, titled “Growth of the Basic Premium: OLS and 2SLS Regressions.” The table estimates the relationship between the concentration of Low Income Premium Subsidy Amount weights and growth in the average basic premium in Medicare Part D markets.

2 Research Question and Motivation

The paper asks whether private insurers strategically responded to the design of the Low Income Subsidy in Medicare Part D. The central concern is that the low income premium subsidy amount, or LIPSA, is calculated using plan premiums. Since many insurers offer multiple plans, the subsidy formula may allow insurers to influence the subsidy threshold through their pricing decisions.

This question is economically and politically important because Medicare Part D is a major public insurance program delivered through private insurers. If the subsidy design encourages strategic premium manipulation, then government spending can rise without a corresponding improvement in consumer welfare.

The principal contribution of the paper is to show that the design of a public subsidy can create strategic pricing incentives for private insurers. The paper combines plan-level descriptive evidence with market-level regressions and an instrumental variables strategy to estimate whether more manipulable subsidy weights are associated with faster premium growth.

3 Data

This replication uses the author-provided file `master_data_file.dta`. The original replication materials describe this file as the master dataset used for the analysis. The original data can be reconstructed from public CMS, Q1Medicare, formulary, and IPUMS sources, but this replication uses the prepared author replication dataset.

The Python script expects the data file to be located at:

```
data/master_data_file.dta
```

The final regression dataset used for Table 8 is at the Medicare Part D region-year level. There are 34 geographic Medicare Part D regions and five regression years, 2007 through 2011, giving 170 region-year observations.

The dependent variable is the change in the log of the enrollment-weighted average basic premium. The main explanatory variable is $wLIS4$, which is the sum of the four largest LIPSA weights among basic Prescription Drug Plans in a region-year. This variable measures the concentration of LIPSA weights and therefore proxies for how manipulable the subsidy may be.

The main instrument is the Medicare Advantage Prescription Drug plan enrollment share in 2006 interacted with an indicator for years after 2008. The paper uses this instrument because the LIPSA weighting rule changed in 2009, and regions with different pre-existing MA-PD penetration were affected differently by that national rule change.

The submitted repository does not include the `master_data_file.dta` dataset. Instead, the `data/README.md` file explains where the dataset should be placed before running the code.

4 Empirical Strategy

The main estimating equation is:

$$\Delta \ln(\text{basic premium})_{jt} = \alpha + \beta wLIS4_{jt} + \gamma X_{jt} + \delta Q_{j,t-1} + \tau_t + \lambda_j + \varepsilon_{jt}, \quad (1)$$

where j indexes Medicare Part D regions and t indexes years. The coefficient of interest is β . A positive value of β means that higher concentration of LIPSA weights is associated with faster growth in basic premiums.

The paper estimates this relationship using both ordinary least squares and two-stage least squares. The instrumental variables approach is used because $wLIS4$ is mechanically

related to premiums and enrollment weights. This mechanical relationship could bias OLS estimates.

The excluded instrument is:

$$MA-PD\ Share_{j,2006} \times Post2009_t. \quad (2)$$

The identifying assumption is that, after controlling for region fixed effects, year fixed effects, controls, and in some specifications region-specific time trends, the instrument affects basic premium growth only through its effect on LIPSA weight concentration.

Standard errors are clustered by region, matching the inference procedure used in the original paper.

5 Replication Process

The replication was implemented in Python. The Python code ports the relevant logic from the original Stata workflow into a reproducible Python script.

The main steps are:

1. Load `data/master_data_file.dta`.
2. Construct basic PDP indicators and LIPSA eligibility variables.
3. Recreate the region-year market-level dataset.
4. Compute the enrollment-weighted average basic premium.
5. Construct the dependent variable, $\Delta \ln(\text{basic premium})$.
6. Construct `wLIS4`, the sum of the four largest LIPSA weights in each region-year.
7. Construct the 2006 MA-PD share interacted with the post-2008 indicator.
8. Estimate the six OLS specifications and six 2SLS specifications from Table 8.
9. Save the output tables to `output/tables/`.

The script saves three output files:

```
output/table4_panel_c_market_summary.csv
output/table8_key_coefficients.csv
output/table8_replication.csv
```

This is an approximate software replication because the original analysis was written in Stata, while this replication uses Python. The main coefficient estimates and clustered standard errors for *wLIS4* match the published Table 8 up to rounding.

6 Replication Results

Table 1 reports the key replicated coefficients on *wLIS4*. The estimates are positive across both OLS and 2SLS specifications. This reproduces the main conclusion of Table 8: regions with more concentrated LIPSA weights experienced faster growth in basic premiums.

Table 1: Key Replicated Coefficients from Decarolis Table 8

	(1)	(2)	(3)	(4)	(5)	(6)
OLS coefficient on <i>wLIS4</i>	0.274	0.301	0.276	0.302	0.304	0.352
Clustered SE	0.059	0.150	0.075	0.156	0.077	0.143
2SLS coefficient on <i>wLIS4</i>	0.544	0.616	0.689	0.632	0.750	0.702
Clustered SE	0.237	0.324	0.242	0.336	0.252	0.314
Observations	170	170	170	170	170	170

Notes: The table reports the main coefficient of interest from the replicated Table 8. Standard errors are clustered by Medicare Part D region. The six columns correspond to the six model specifications in the original Table 8.

In the simplest OLS specification, the coefficient on *wLIS4* is 0.274 with a clustered standard error of 0.059. In the simplest 2SLS specification, the coefficient is 0.544 with a clustered standard error of 0.237. The 2SLS estimates are larger than the OLS estimates, consistent with the paper’s concern that OLS may be biased by the mechanical relationship between premiums and LIPSA weights.

Economically, the positive estimates suggest that greater manipulability of the LIPSA formula is associated with faster premium growth. This supports the paper’s argument that the Low Income Subsidy design created incentives that distorted insurer pricing.

7 Differences and Limitations

The replicated coefficients and clustered standard errors for the main variable match the published Table 8 up to rounding. Some constant terms, especially in specifications with region-specific time trends, may differ because Python and Stata can handle fixed effects, omitted categories, and collinear time-trend variables differently. These differences do not affect the main coefficient of interest, *wLIS4*.

A second limitation is that this replication uses the author-provided `master_data_file.dta` rather than reconstructing the dataset from the original CMS, Q1Medicare, formulary, and IPUMS source files. Therefore, the replication verifies the Table 8 analysis from the prepared author dataset, but it does not independently rebuild the full raw dataset.

Overall, the main economic conclusion is reproduced: higher concentration of LIPSA weights is associated with faster growth in Medicare Part D basic premiums.

8 Generative AI Disclosure

ChatGPT and Gemini were used for coding support, workflow planning, debugging, and documentation. Specifically, it was used to help interpret the replication materials, translate the relevant Stata workflow into Python, debug the construction of the region-year dataset, format regression outputs, and draft the replication report and README documentation.

The submitting student(s) reviewed the final code and report, verified the reported results against the Python output, and remain responsible for understanding and explaining all submitted work.

A Full Replicated Table 8 Output

Table 2 reports the full table-like output produced by the Python script and saved as `output/table8_replication.csv`.

B References

References

Decarolis, F. (2015). Medicare part d: Are insurers gaming the low income subsidy design? *American Economic Review*, 105(4):1547–1580.

Table 2: Replication of Decarolis Table 8: Growth of the Basic Premium

Variable	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. OLS</i>						
<i>wLIS4</i>	0.274*** [0.059]	0.301* [0.150]	0.276*** [0.075]	0.302* [0.156]	0.304*** [0.077]	0.352** [0.143]
HHI			0.155 [0.469]	0.187 [0.531]	0.088 [0.471]	-0.066 [0.735]
Unemployment			-0.0077 [0.011]	-0.028 [0.027]	-0.0072 [0.011]	-0.023 [0.025]
Plan age					-0.071 [0.067]	-0.054 [0.104]
Pharmacies					-0.176 [1.313]	-1.033 [1.636]
Drugs					-0.292 [0.356]	-0.293 [0.484]
Constant	-0.110*** [0.014]	-0.047 [0.031]	-0.114 [0.075]	0.048 [0.113]	0.239 [0.348]	0.363 [0.455]
Region time trends	No	Yes	No	Yes	No	Yes
R-squared	0.472	0.636	0.476	0.644	0.488	0.654
Observations	170	170	170	170	170	170
<i>Panel B. 2SLS</i>						
<i>wLIS4</i>	0.544** [0.237]	0.616* [0.324]	0.689*** [0.242]	0.632* [0.336]	0.750*** [0.252]	0.702** [0.314]
HHI			-0.378 [0.565]	-0.177 [0.526]	-0.498 [0.584]	-0.647 [0.686]
Unemployment			-0.014 [0.013]	-0.030 [0.029]	-0.014 [0.013]	-0.022 [0.028]
Plan age					-0.107* [0.063]	-0.061 [0.108]
Pharmacies					-0.200 [1.466]	-1.198 [1.614]
Drugs					-0.412 [0.485]	-0.521 [0.502]
Constant	-0.165*** [0.049]	0.016 [0.067]	-0.071 [0.087]	0.154 [0.164]	0.433 [0.468]	0.685 [0.485]
Region time trends	No	Yes	No	Yes	No	Yes
R-squared	0.433	0.609	0.401	0.617	0.403	0.625
Observations	170	170	170	170	170	170

Notes: Standard errors clustered by Medicare Part D region are reported in brackets. All specifications include region and year fixed effects. Pharmacies is scaled by 10,000 for readability. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.